

ME 599 J, K / IND E 599 A
AI/ML for Smart Manufacturing

Description: Data-driven statistical clustering, dimensionality reduction, classification, and regression methods; deep neural networks, Bayesian networks; robot intelligence; for the design, development, and operation of smart manufacturing systems.

Objectives: This course is designed to educate graduate students on the use of AI and ML tools and techniques to address various challenges with smart manufacturing, such as complex parts design and materials selection, process monitoring and control, quality inspection, preventive maintenance, yield/throughput optimization, and supply chain management. It will cover some of the prevalent as well as emerging AI/ML methods for both predictive and prescriptive analytics, including the use of embodied agents, such as collaborative robots.

Instructors: Professor Ashis G. Banerjee
Email: ashisb@uw.edu
Office Hour: T 2:30 – 3:30 pm; SIG 219 or
<https://washington.zoom.us/j/94781670526>

Professor Krithika Manohar
Email: kmanohar@uw.edu
Office Hour: Th 2:30 – 3:30 pm; MEB 209

Lectures: M 4:30 pm - 5:50 pm SIG 433
W 4:30 pm – 5:50 pm SIG 433

Textbook: None

Web Site: <https://canvas.uw.edu/courses/1890126>

Format: **Lectures:** 2 eighty-minute lectures every week

Homework Assignments: There will be four homework assignments. They will be assigned on the 2nd, 4th, 6th, and 9th weeks, respectively, and will be due ten to twelve days from the assigned dates. Each assignment will involve manufacturing problem identification, AI/ML model selection, development, and implementation through computer programming. Sample source code in a language known to everyone may be provided to assist with programming. You can discuss the homework assignments with your classmates (or use online resources, including AI tools), but need to submit your own independently typed solutions. No late

submissions will be allowed without explicit prior permission of the instructor owing to genuine and unavoidable circumstances.

Project: There will be a quarter-long project to replicate the AI/ML outcomes on an already published (preferably, with open access data) manufacturing problem. A list of representative topics will be provided in the class, and the topic should be selected after consulting the instructor by the middle of the 2nd week of class. You can also choose a topic based on your own research work, but then the onus will be on you to demonstrate the validity of the outcomes. The project should be done in groups of two.

Requirements involve mid-term and final project presentations, and a detailed project report in the style of an IEEE Conference Proceedings article. Weekly updates, presentations, and report formats will be discussed in the class.

Class assignments: Each code demonstration will be accompanied by a short assignment, due on the same day via Canvas.

Miscellaneous: Each project group will get access to a UW Google shared drive folder. You can submit your project materials in your respective folders as well as use them for coordination purposes.

All the lecture materials, including relevant papers, data sets, videos, etc. will be uploaded to the Canvas course site just before they are presented in the class but may be revised after the class is over. All the homework assignments and exam questions (and corresponding grades) will also be available through the Canvas course site. You should receive automatic notifications from Canvas whenever any homework/exam questions are posted.

Grading: Homework assignments: 25%
Take-home exam: 20%
Class assignments: 20%

Project: 35% (10% for mid-term presentation, 10% for end-term presentation, and 15% for report)

Grading policy: Course grading will be relative and based on percentile scores. No one is expected to be awarded a grade below 3.0 unless that person significantly underperforms with respect to the rest of the class.

The take-home exam must be done independently without any consultation or discussion with your course colleagues. Feel free to contact the instructor though at any point of time with questions or clarifications. Any instance of cheating or other form of Student Honor Code violation will be dealt with firmly. It is likely to

result in a penalty of *zero grade* for that particular assignment and will be reported to the UW COE Academic Affairs Committee.

Email List: Your UW emails will be used to communicate with you individually. Class notifications will be announced on Canvas.

Disability Accommodation: Contact Disability Resources for Students, 011 Mary Gates, 543-8924 (V/TTY) and discuss with your instructor.

Religious Accommodation: Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including more information about how to request an accommodation, is available at Religious Accommodations Policy (<https://registrar.washington.edu/staffandfaculty/religious-accommodations-policy/>). Accommodations must be requested within the first two weeks of this course using the Religious Accommodations Request form (<https://registrar.washington.edu/students/religious-accommodations-request/>).

Success Tips: Attend all the lectures and be actively engaged. *Take notes* as anything that gets discussed in the class, including the presented papers, will be considered as content for the homework assignments and exams. *Watch the additional videos* on catch-up lecture content and/or explanations or clarifications of lecture concepts or assigned problems. Please *make good use of the office hours* to have a better understanding of how to solve the homework problems, tackle the exam questions, and execute the project well.

Do not leave the work until right before the submission deadlines. Spend at least a couple of hours working on the project every week. *Show all the steps and explain the rationale* behind your answers in detail for homework and exam questions. Many of the questions will require substantial literature search, analytical thinking, and computational work.

The course will assume knowledge of undergraduate-level linear algebra, applied statistics, manufacturing processes or systems, and computer programming.

Schedule

Week	Date	Topics
1	3/30 4/1	Course overview and introduction; dimensionality reduction Dimensionality reduction

2	4/6 4/8	Clustering; Project topics and format discussed Clustering; HW # 1 assigned; Project groups finalized
3	4/13 4/15	Classification Classification
4	4/20 4/22	Classification; HW # 1 due Classification; HW # 2 assigned
5	4/27 4/29	Regression Regression; Midterm project presentations
6	5/4 5/6	Bayesian regression; HW #2 due; HW # 3 assigned Bayesian regression;
7	5/11 5/13	Neural networks Neural networks with applications in machine vision; HW # 3 due; Take-home exam assigned
8	5/18 5/20	Robotics applications Robotics applications
9	5/25 5/27	Emerging topics (generative AI); Take-home exam due Emerging topics; HW # 4 assigned
10	6/1 6/3	Final project presentations Final project presentations
11	6/8 6/10	HW # 4 due Project reports due